

Convert the CFG to CNF.

$S \rightarrow aAD$

$A \rightarrow aB \mid bAB$ } Not in CNF.

$\checkmark B \rightarrow b$
 $\checkmark B \rightarrow d$ } in CNF

Soln

(1) Eliminate unit productions.

(2) $\xrightarrow{\quad}$ Epsilon $\xrightarrow{\quad}$

(3) $\xrightarrow{\quad}$ useless symbol. $\xrightarrow{\quad}$ No useless

Here

$A \rightarrow B$ — No unit

$A \rightarrow \epsilon$ — No epsilon

? If $\boxed{\begin{matrix} A \rightarrow BC \\ A \rightarrow a \end{matrix}}$ Condition for CNF.

Nonterminal $\rightarrow$ Exactly two NT

Non terminal $\rightarrow$ Single terminal

Here

for. $S \rightarrow aAD$

$\checkmark S \rightarrow CaD_1$

$C_a \rightarrow a$

$\checkmark D_1 \rightarrow AD$

So CNF is.

for $A \rightarrow aB$

$\checkmark A \rightarrow CaB$

$\checkmark C_a \rightarrow a$

for $A \rightarrow bAB$

$\checkmark A \rightarrow C_b D_2$

$\checkmark D_2 \rightarrow AB$

$\checkmark C_b \rightarrow b$

$S \rightarrow CaD_1$
 $D_1 \rightarrow AD$
 $A \rightarrow CaB \mid C_b D_2$
 $C_a \rightarrow a$
 $C_b \rightarrow b$
 $B \rightarrow b \mid d$
 $D_2 \rightarrow AB$

Consider the following ^{CFG} Grammar convert it into ~~CFG~~ ^{Chomsky Normal form}.

$$S \rightarrow aAbB$$

$$A \rightarrow Ab|b$$

$$B \rightarrow Ba|a.$$

In CNF

$$A \rightarrow BC \text{ (two Non-terminal)}$$

$$A \rightarrow a \text{ (single terminal)}$$

Soln

Step 1

Introduce new Variable

$$X \rightarrow a$$

$$Y \rightarrow b.$$

New Rewrite

$$S \rightarrow XAYB$$

$$A \rightarrow AY|b$$

$$B \rightarrow BX|a.$$

$$X \rightarrow a$$

$$Y \rightarrow b$$

Step 2 — Break long production into binary form.

$$S \rightarrow XAYB \text{ (break it)}$$

$$\text{Let } S \rightarrow XS_1$$

$$S_1 \rightarrow AS_2$$

$$S_2 \rightarrow YB$$

Step 3 final CNF Grammar is.

Variable are S, S_1, S_2, A, B, X, Y .

$$S \rightarrow XS_1$$

$$S_1 \rightarrow AS_2$$

$$S_2 \rightarrow YB.$$

$$A \rightarrow AY|b$$

$$B \rightarrow BX|a.$$

$$X \rightarrow a$$

$$Y \rightarrow b.$$

Convert the following grammar into CNF

$$S \rightarrow ASA \mid bA$$

$$A \rightarrow B \mid S \leftarrow \text{unit production.}$$

$$B \rightarrow c \quad (\text{terminal})$$

Soln

Step 1

Eliminate unit production.

Unit productions are:

$$A \rightarrow B \in$$

$$A \rightarrow S.$$

$$W_0(A) = \{A, B, S\}$$

$$W_1(A) = W_0(A) \cup \phi$$

Since $B \rightarrow c$

$$S \rightarrow ASA \mid bA.$$

So $A \rightarrow B$ become.

$$A \rightarrow c \mid ASA \mid bA.$$

Updated grammar will be.

$$S \rightarrow ASA \mid \underline{bA}$$

$$A \rightarrow c \mid ASA \mid \underline{bA}.$$

$$B \rightarrow c$$

Step 2 Replace $X \rightarrow b$ so our grammar become.

$$S \rightarrow ASA \mid \underline{XA}$$

$$A \rightarrow c \mid ASA \mid \underline{XA}$$

$$B \rightarrow c$$

$$X \rightarrow b.$$

Step 3

Convert long productions into binary form.

for $S \rightarrow ASA$

$$S \rightarrow AS_1$$

$$S_1 \rightarrow SA$$

Similarly

for $A \rightarrow ASA$

$$A \rightarrow AS_1$$

$$S_1 \rightarrow SA.$$

Step 4 So final grammar will be.

$$S \rightarrow AS_1 \mid \underline{XA}$$

$$S_1 \rightarrow SA$$

$$A \rightarrow AS_1 \mid \underline{XA} \mid c$$

$$B \rightarrow c$$

$$X \rightarrow b$$

} in CNF.

Convert the given CFG into CNF.

$$E \rightarrow E + E \mid E * E \mid a \mid b \mid c.$$

Soln

Step 1 - Replace terminals in mixed productions.

Introduce New Variable

$$P \rightarrow +$$

$$M \rightarrow *$$

Now Rewrite our grammar given.

$$E \rightarrow EPE \mid EME \mid a \mid b \mid c.$$

Step 2 Convert long productions into binary form.

for $E \rightarrow EPE$

$$E \rightarrow EX_1$$

$$X_1 \rightarrow PE$$

for $E \rightarrow EME$

Similarly $E \rightarrow EX_2$

$$X_2 \rightarrow ME$$

Step 3 final CNF is.

$$E \rightarrow EX_1 \mid EX_2 \mid a \mid b \mid c$$

$$X_1 \rightarrow PE$$

$$X_2 \rightarrow ME$$

$$P \rightarrow +$$

$$M \rightarrow *$$

CNF.

Convert the given CFG into CNF.

Soln

$$E \rightarrow E + E \mid E * E \mid a \mid b \mid c.$$

Step 1 - Replace terminals in mixed productions.

Introduce New Variable

$$P \rightarrow +$$

$$M \rightarrow *$$

Now Rewrite our grammar given.

$$E \rightarrow EPE \mid EME \mid a \mid b \mid c.$$

Step 2 Convert long productions into binary form.

for $E \rightarrow EPE$

$$E \rightarrow EX_1$$

$$X_1 \rightarrow PE$$

for $E \rightarrow EME$

similarly $E \rightarrow EX_2$

$$X_2 \rightarrow ME$$

Step 3 final CNF is.

$$E \rightarrow EX_1 \mid EX_2 \mid a \mid b \mid c$$

$$X_1 \rightarrow PE$$

$$X_2 \rightarrow ME$$

$$P \rightarrow +$$

$$M \rightarrow *$$

CNF.

is said to be in GNF if

Convert the following CFG into CNF.

$S \rightarrow aAa | bBb$

$A \rightarrow C | a$

$B \rightarrow C | b$

$C \rightarrow CDE | \epsilon$

$D \rightarrow A | B | ab$

$C \rightarrow \epsilon$ so $w_1 = \{C\}$
as $B \rightarrow C$ & $A \rightarrow C$ & $C \rightarrow \epsilon$
so $w_2 = \{C, A, B\}$.

$w_1(n) \rightarrow \{D, B, A\}$ unit
 $D \rightarrow a | b | ab$ production

- ① ϵ
② unit
③ useless

Soln

Step 1 - Remove ϵ -production.

$C \rightarrow \epsilon$ so C is nullable

Now remove C from all production

$A \rightarrow C | a$ it become $A \rightarrow a$

$B \rightarrow C | b$ it become $B \rightarrow b$

$C \rightarrow CDE$ it become $C \rightarrow CDE | DE$

So updated Grammar will be.

$S \rightarrow aAa | bBb$

$A \rightarrow a$

$B \rightarrow b$

$C \rightarrow CDE | DE$

$D \rightarrow A | B | ab$

So new Production.
 $S \rightarrow aAa | bBb$

$A \rightarrow a$

$B \rightarrow b$

$C \rightarrow CDE | DE$

$D \rightarrow a | b | ab$

Step 2

Remove unit production

Replace $D \rightarrow A \rightarrow a$
 $D \rightarrow B \rightarrow b$

$D \rightarrow A | B$

so $D \rightarrow a | b | ab$

Step-3 Replace terminals in long productions.
introduce New Variable.

$X \rightarrow a$

$Y \rightarrow b$

So

$S \rightarrow XAX | YBY$

$D \rightarrow a | b | XY$

Step 4 break $S \rightarrow XBY$

$S \rightarrow YS_2$ & $S_2 \rightarrow BY$

break $C \rightarrow CDE$ into

$C \rightarrow CS_1$ & $C_1 \rightarrow DE$

Step 5 final Grammar is

$S \rightarrow XS_1 | YS_2$ & $C \rightarrow CC_1 | DE$

$S_1 \rightarrow AX$

$S_2 \rightarrow BY$

$A \rightarrow a$
 $B \rightarrow b$

$C_1 \rightarrow DE$

$D \rightarrow a | b | XY$

$X \rightarrow a$ & $Y \rightarrow b$

Step 4 Convert to binary

Break $S \rightarrow XAX$

so $S \rightarrow XS_1$ & $S_1 \rightarrow AX$

Convert the following CFG to CNF

$$S \rightarrow aAD$$

$$A \rightarrow aB \mid bAB$$

$$B \rightarrow b$$

$$D \rightarrow d.$$

Solⁿ

Step 1 Replace terminal in mixed production.

Introduce New Variable

$$X \rightarrow a \quad \& \quad Y \rightarrow b$$

Now rewrite

$$S \rightarrow XAD$$

$$A \rightarrow XB \mid YAB$$

$$B \rightarrow b$$

$$D \rightarrow d.$$

$$X \rightarrow a$$

$$Y \rightarrow b$$

Step 2 - Convert the long production into binary form.

$S \rightarrow XAD$ can be written as $\rightarrow$

$$S \rightarrow XS_1$$

$$S_1 \rightarrow AD.$$

$A \rightarrow YAB$ can be written as $\rightarrow$

$$A \rightarrow YA_1$$

$$A_1 \rightarrow AB.$$

Step 3 final grammar which is CNF.

$$S \rightarrow XS_1$$

$$S_1 \rightarrow AD.$$

$$A \rightarrow XB \mid YA_1$$

$$A_1 \rightarrow AB.$$

$$B \rightarrow b$$

$$D \rightarrow d.$$

$$X \rightarrow a$$

$$Y \rightarrow b.$$

Reduce the below grammar into Chomsky Normal form

$$S \rightarrow 0S0 \mid 1S1 \mid A$$

$$A \rightarrow 2B3$$

$$B \rightarrow 2B3 \mid 3$$

Soln

Step 1 Remove unit production.

$S \rightarrow A$ Replace A by $2B3$.

The grammar become.

$$S \rightarrow 0S0 \mid 1S1 \mid 2B3$$

$$A \rightarrow 2B3$$

$$B \rightarrow 2B3 \mid 3$$

Step 2 - Replace terminal in mixed production.

i.e. $x_0 \rightarrow 0$, $x_1 \rightarrow 1$, $x_2 \rightarrow 2$, $x_3 \rightarrow 3$

$\hookrightarrow$ Rewrite grammar.

$$S \rightarrow x_0 S x_0 \mid x_1 S x_1 \mid 2B3 \mid x_2 B x_3$$

$$A \rightarrow x_2 B x_3$$

$$B \rightarrow x_2 B x_3 \mid 3$$

Step 3 $\rightarrow$ Convert long productions into binary form.

Break $S \rightarrow x_0 S x_0$ into $S \rightarrow x_0 S_1$
 $S_1 \rightarrow S x_0$

Break $S \rightarrow x_1 S x_1$ into $S \rightarrow x_1 S_2$
 $S_2 \rightarrow S x_1$

Break $S \rightarrow x_2 B x_3$ into $S \rightarrow x_2 S_3$
 $S_3 \rightarrow B x_3$

Break $A \rightarrow x_2 B x_3$ into $A \rightarrow x_2 A_1$

Break $B \rightarrow x_2 B x_3$ into $B \rightarrow x_2 B_1$
 $B_1 \rightarrow B x_3$

So final grammar is

$$\begin{array}{l} S \rightarrow x_0 S_1 \mid x_1 S_2 \mid x_2 S_3 \\ S_1 \rightarrow S x_0 \\ S_2 \rightarrow S x_1 \end{array} \quad \begin{array}{l} S_3 \rightarrow B x_3 \\ A \rightarrow x_2 A_1 \\ A_1 \rightarrow B x_3 \end{array} \quad \begin{array}{l} B \rightarrow x_2 B_1 \mid 3 \\ B_1 \rightarrow B x_3 \end{array} \quad \begin{array}{l} x_0 \rightarrow 0 \quad x_2 \rightarrow 2 \\ x_1 \rightarrow 1 \quad x_3 \rightarrow 3 \end{array}$$

back Normal form (GNF)

Reduce the below grammar into chomsky Normal form

$$G = (\{S\}, \{a, b, c\}, \{S \rightarrow a|b|cSS\}, \{S\})$$

Soln The given $S \rightarrow a|b|cSS$

Step 1 Replace terminal in mixed production.

$$\text{i.e. } C \rightarrow c$$

$$\text{so } S \rightarrow a|b|cSS$$

Step 2 - Convert long production into binary form.

$$S \rightarrow CSS \text{ into } \begin{aligned} S &\rightarrow CS_1 \\ S_1 &\rightarrow SS \end{aligned}$$

Step 3 - final CFG.

$S \rightarrow a b CS_1$
$S_1 \rightarrow SS$
$C \rightarrow c$